SPECIAL PACKAGING INSTRUCTION

Form Approved *-AM*
OMB No. 0704-0188 *JO*

1. PART OR DRAWING NO. (CAGE) NOMENCLATURE	2. CAGE	3. SPI NO.
M52109 (19207) PUMP UNIT,CENTRIFUGAL	19207	AK16911985

4. NATIONAL STOCK NO.	5. DATE	6. REVISION
4320-01-691-1985	07-17-2024	

7. QUP	8. ICQ	9. UNIT PACK WT (LB.)	10. UNIT PACK CU (CU. FT.)	11. UNIT PACK SIZE (INCHES)
001	NONE	319.0	19.657	36.3 x 29.8 x 31.4

	18. STEPS	19. REQD	20. DESCRIPTION
12. MILITARY PRESERVATION MIL-STD-2073-1 METHOD 42			
13. CLEANING MIL-STD-2073-1	A		Preservation for interior surfaces
	B		Preservation for exterior unpainted metals
14. DRYING MIL-STD-2073-1	C		Preserve engine according to ATPD-2232, Type II, Method 20
	D	1	Container: ASTM D 6251, Type III, Class 2, Style A, Treatment A
15. PACKING			
a. LEVEL A MIL-STD-2073-1 AND NOTE E			
b. LEVEL B NOT APPLICABLE			
16. MARKING MIL-STD-129			

17. NOTES/DRAWING

A. Drain and dry pump. Coat interior surfaces with DOD-L-24651 or USDA/NSF 60 class H3 or H1 preservative.
B. Coat unpainted metal surfaces with MIL-PRF-16173, Grade I preservative.
C. The engine and engine accessories shall be preserved in accordance with the level A requirements of ATPD-2232, as specified for Type II, Method 20.
D. Deck Assembly: (See accompanying illustrations).

1. Skids are attached to 3/4 in. CDX plywood pallet IAW ASTMD 6251
2. Registers (2 x 4 s) are attached (edge up) to floater (3/4 in. CDX plywood 31” X 23”)
3. Floater is centered on plywood base.
4. Holes are drilled through the top of the registers, through the floater deck board and through the pallet base 6” from the end on the long side of the floating base.
5. Four strips of 2 in. x 2 in., 2# closed cell foam (Spec: A-A-59136, Class 2, Grade A, Type I) are then glued and centered on the 3/4” pallet base.
6. Carriage bolts (3/8 – 16 X 9 in.) are placed through the bottom of the 3/4 in. pallet base, through the floater, and then out the top of the registers.
7. Compression type rebound springs (high carbon steel, zinc plated, length: 1.25 in; O.D. .75 in.; I.D. .50 in.; travel ..50 in.) are then placed over the top of the protruding carriage bolt and secured in place by a flat washer, lock washer, and lock nuts with secured with a commercial grade of “ND Industries” or “Loc-Tite” thread-lock.
8. The panel sides and ends (3/8” CDX & 1 x 4s) are then fastened together IAW ASTM D 6251.
9. The pallet base assembly, with the floater attached, is then placed upside down on the assembled panel sides and ends and attached IAW ASTM D6251.
10. The crate/box assembly is then set upright.
11. A vapor barrier bag MIL-DTL-117, Type I, Class E, Style 1 is then placed on the outside of the corrugated container (See ISO drawing) and placed inside the wood crate with the box securely between the registers.
12. Double wall 1/4” corrugated inserts are then placed between the rebound springs and vapor barrier bag to prevent damage to the vapor barrier bags by the bolt/spring assembly.

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17. NOTES/DRAWINGS *CONTINUED*

13. The pump unit is then lifted by its handles and placed within the corrugated box (which is within the vapor barrier bag) sealed securely between the registers. No internal blocking or bracing of the pump is needed. The tight fit of the pump within the carton and inside the registers minimizes side to side movement; and steps 20, 22, and 23 below securely hold the pump in place by banding and tightening diagonally across the pump handles located along top and full length of and on both sides of the pump unit.
14. Place the manual in a waterproof, heat sealable bag, Size: 9” x 11-1/2” and heat seal closed. Secure bagged manual to pump with tie wrap.
15. Close corrugated container IAW ASTM-D 1974 Method 2B3.
16. The vapor barrier bag (MIL – DTL-117, Type I, Class E, Style 1) is then partially sealed with a heat sealer.
17. Through the unsealed portion of the barrier bag (approximately 2-4 in.) as much air as practical is then removed from the barrier bag by use of a vacuum cleaner or hand pressure until the bag conforms to the shape of the corrugated container.
18. The vapor barrier bag is then sealed closed with the heat sealer.
19. Top of the vapor barrier bag is then folded and taped in place.
20. Metal strapping material is then placed under the floating deck and around the vapor barrier bag containing the pump within the corrugated container.
21. Protector pads are then placed on the edges of the barrier bag covered corrugated container to protect the vapor barrier bag from the metal bands.
22. Metal bands are then tightened to hold the packaged unit securely in place.
23. Close container IAW ASTM D6251.
24. Steel strapping is then placed around the wooden crate IAW ASTM D 6251.

E. The unit container is the shipping container.

F. Quality Assurance Provisions:
1. Inspect the preservation and unit pack in accordance with MIL-STD-2073-1
2. Quality conformance shall be in accordance with MIL-STD-2073-1.

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